HBS 9-119-020 and 9-119-021 . Datar and Bowler (2018)

LendingClub (A) and (B)

Data Analytic Thinking . Decision Trees . Random Forests

How Emily Figel built a default-prediction model on 413,000 peer-to-peer loans - and which model finally worked.

Presented by

Group 6

PGDM-BM and PGDM-BA . MDI Gurgaon

Course

Strategy Formulation and Implementation

Prof. Sangeeta Shah Bharadwaj . Term IV (Jun-Aug 2026)

ROADMAP FOR TODAY

Agenda

01 **The setup**

Emily Figel and the question her firm is asking

02 **How LendingClub works**

The business model, grades, and the $10,000 example

03 **The data**

413,000 loans, the target variable, and one paradox

04 **Building Model 1 (Case A)**

Cross-validation, data leakage, and a realistic second attempt

05 **The risks around the model**

Borrower quality, transparency, and LC as a counterparty

06 **Trees and Forests (Case B)**

Decision trees, Gini, pruning, and Random Forest

07 **Model showdown and recommendation**

What the firm should actually do

Meet Emily Figel and her first client-facing question

THE PROTAGONIST

Emily Figel

. MBA student, joining a Chicago wealth-management firm in the fall

. Fascinated by alternative banking models

. Advises aggressive clients where small mistakes scale fast

HER FIRST PROJECT

Is LendingClub a reasonable investment vehicle for the firm's aggressive clients?

HER PLAN . WHAT SHE ACTUALLY WANTS TO DO

1. **Learn the basics** of predictive modelling using LendingClub's own historical data
2. **Get a feel** for the opportunities and risks of investing on the platform
3. **Build a classifier** to predict whether a new borrower will default or repay
4. **Use predictions** as one input among several before funding any loan

How LendingClub actually works (not really peer-to-peer)

2007

Platform launched (one of the first P2P lenders alongside Zopa, Prosper)

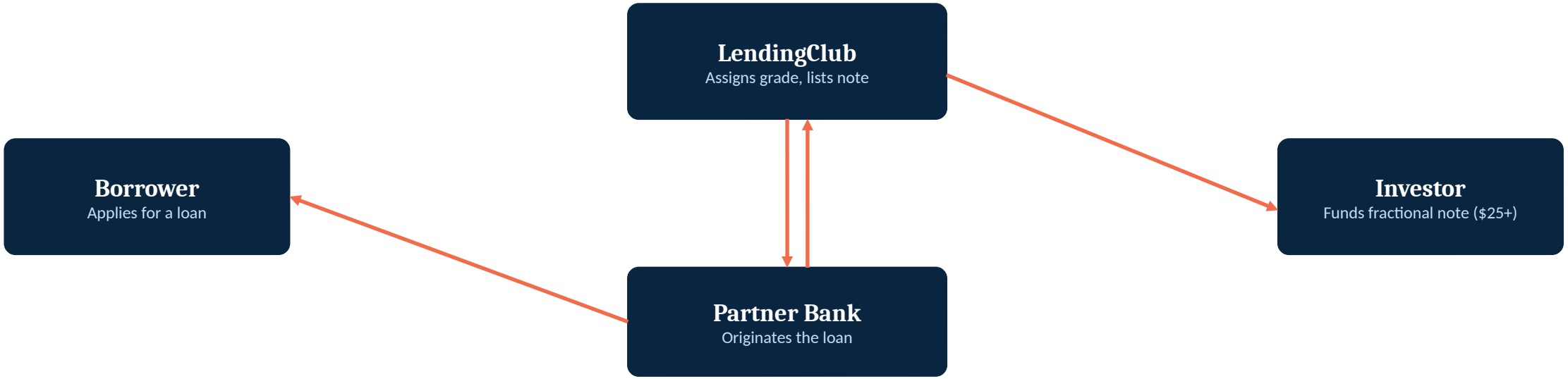
Dec 2014

Largest tech IPO of the year; raised nearly $900M

$1.5B / qtr

Loans issued by 2016 (up from $1B cumulative in 2012)

THE LENDING PROCESS (EXHIBIT 2)



Why it matters: the investor never lends directly. She buys a note from LendingClub. If LC files for bankruptcy, she becomes an unsecured creditor with no guaranteed claim on her borrower's payments.

Follow the money: a $10,000 A1 loan at 6.03%

$10,000	3.5%	$9,650	$304.36	$10,956.79
Loan amount	**Origination fee**	**Actually disbursed**	**Monthly payment**	**Total repaid**
Borrower applies, LC grades A1	LC deducts $350 up front	What lands in borrower's account	For 36 months (LC keeps $3.05 as 1% servicing)	$956.79 total interest by month 36

EFFECTIVE INTEREST RATE (not stated rate)

6.24%

effective rate on cash actually received

. Interest paid in Year 1 **$10,000 x 0.0603 = $603**
. Divided by amount disbursed **$603 / $9,650 = 6.24%**

KEY INSIGHT FOR THIS CASE

Interest is front-loaded. 16.5% of the first payment goes to interest; only 0.5% by the last payment.

So even when a borrower defaults, **she has usually paid several months of interest first.**

That is why the platform's **18% default rate** still leaves investors with a **7.4% net return.**

The data: 413,000 loans, 100+ features, one messy label

413K
loans
aggregated from 3 CSV files (2013 to Q1 2015)

100+
features
borrower, credit history, loan terms, LC actions

75K
defaulters (1)
Default + Charged Off

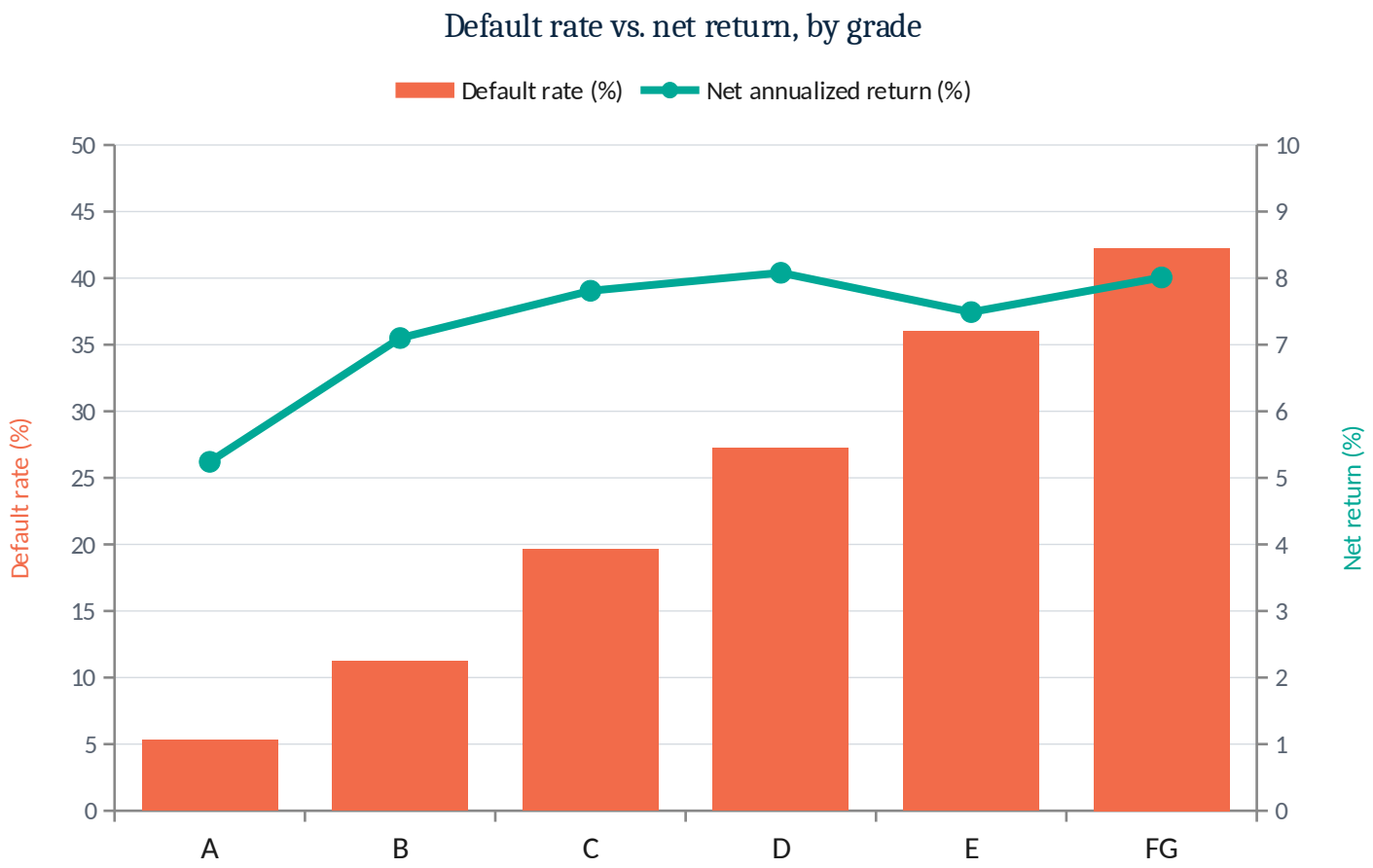
338K
repayers (0)
Fully Paid

DEFINING THE TARGET . TABLE 1 . LOAN STATUS CLASSES

Value	Definition	Target label	Figel's decision
Fully paid	The loan has been repaid in full.	**0**	**Repay**
Current	The loan is still being repaid.	**-**	**Excluded**
Late 16-30 days	Not current for 16-30 days.	**-**	**Excluded**
Late 31-120 days	Not current for 31-120 days.	**-**	**Excluded**
Default	Not paid in 121 days or more.	**1**	**Default**
Charged Off	In default 30+ days, no future payments expected.	**1**	**Default**

'Current' and 'Late' loans are excluded because their eventual outcomes are not observed yet.

The paradox: high default rates, still positive returns



HOW TO READ IT

Default rate climbs sharply, from **5.3% (A) to 42.2% (FG).**

Net return stays nearly flat, **5.2% to 8.1%.**

Overall default rate = 17.97%, net return = 7.39%.

Why? 'Default' is discrete, but payments are continuous. Even defaulted borrowers pay interest and principal for months first.

Investable implication: predicting default/repay is not enough. Figel really cares about expected return by grade. But default classification is the tractable first step.

Splitting 413K loans: training, cross-validation, holdout

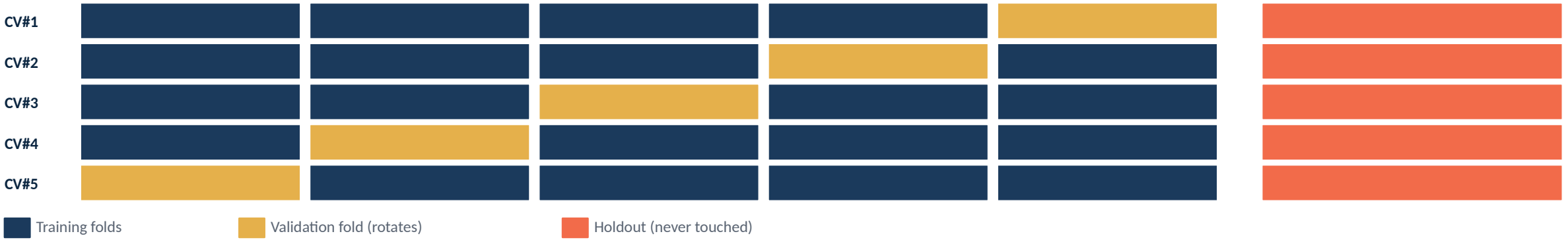
STEP 1 . 80 / 20 RANDOM SPLIT

TRAINING SET . 80% . 330,000 loans (60K defaulters + 270K repayers)	HOLDOUT . 20% . 83K

STEP 2 . TRAINING SET SPLIT INTO 5 EQUAL FOLDS

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	HOLDOUT
66,000 loans	66,000 loans	66,000 loans	66,000 loans	66,000 loans	83,000
12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	*Untouched*

STEP 3 . IN EACH CV RUN, ONE FOLD IS THE VALIDATION SET



Attempt 1 was too perfect. Attempt 2 was believable.

ATTEMPT 1

20 / 20

correct on the 20 obs; probabilities collapsed to 0 or 1; log loss near zero

FIGEL'S HUNCH

"Doubt quickly crowded out my initial sense of awe."

Near-perfect accuracy almost always means: the task is trivial (rarely) or **a feature is leaking the label.**

20 LEAKY FEATURES REMOVED (EXHIBIT 9)

Payment data

loan_status, total_pymnt, total_rec_prncp, total_rec_int, out_prncp, last_pymnt_amnt, pymnt_plan

Post-origination FICO

last_fico_range_high / _low, last_credit_pull_d

Hardship and settlement

hardship_flag / _type / _reason / _status / _amount, debt_settlement_flag, settlement_percentage

Common thread: *these only exist after the loan runs. By construction they 'know' the outcome.*

ATTEMPT 2 (LEAKAGE REMOVED)

16 / 20

directionally correct at a 50% probability cut-off

A GOOD RISK MODEL LOOKS LIKE THIS

8 of 10 defaulters scored P > 47%

8 of 10 repayers scored P < 22%

Probabilities now spread realistically. The model is confident when features are decisive and uncertain when they are not.

Three risks the model cannot see

BORROWER QUALITY

+38%

rise in charge-offs (2013 to 2016)

Income verification collapsed from 49% (2013) to 26.8% (Q1 2016). Nearly half of debt-consolidation borrowers carried 10% more credit-card debt a year later.

TRANSPARENCY

$22M

of loans mislabelled to Jefferies (2016)

Jefferies bought a specific grade level from LC; at least $22M did not meet it, and executives had known. CEO Renaud Laplanche resigned. Trust in the data is the whole business.

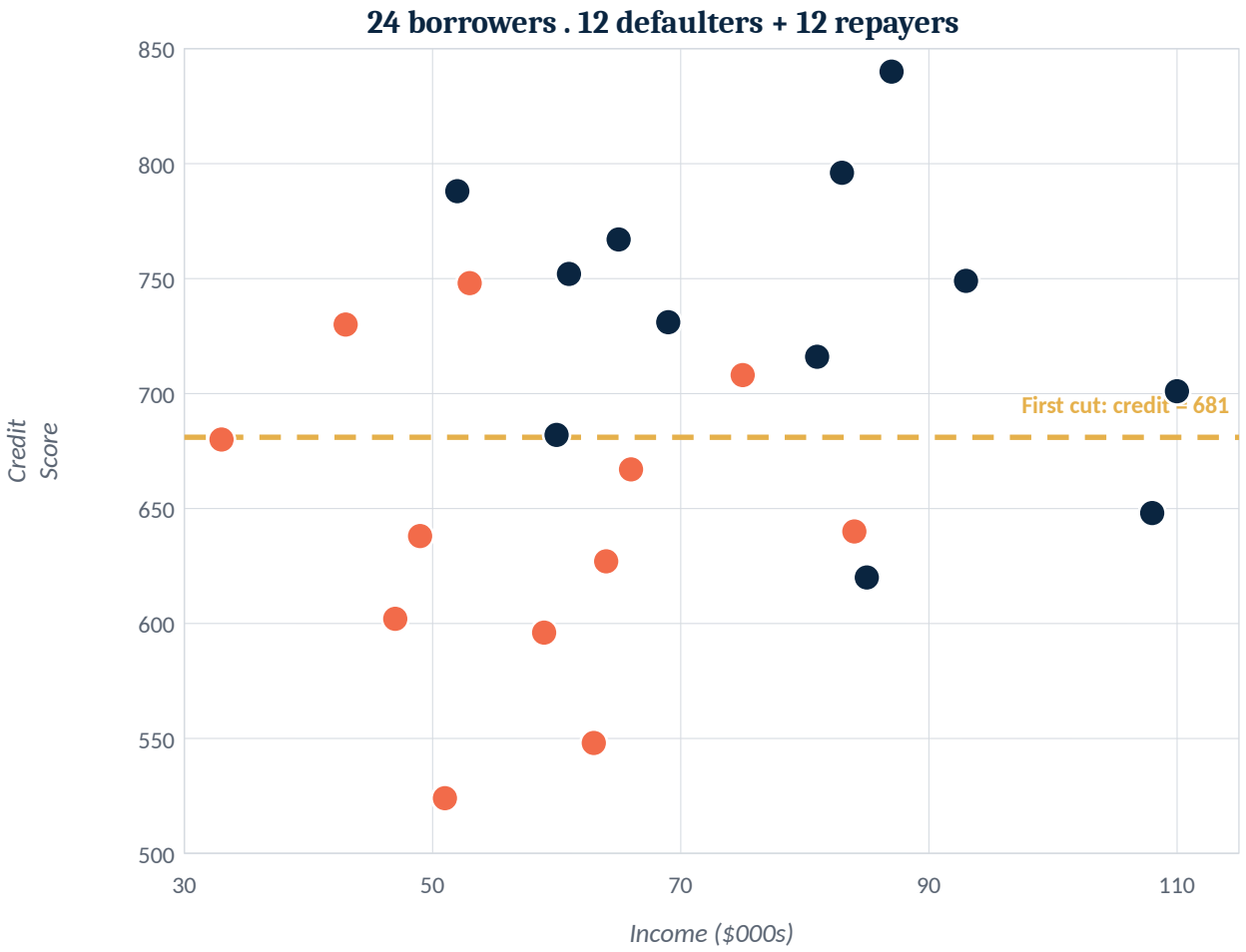
COUNTERPARTY

$0

guaranteed if LC files bankruptcy

The investor holds an unsecured claim against LendingClub, not against her borrower. Even if the borrower keeps paying, cash flows into the LC estate first and she joins the queue.

Decision Trees: the same data as a set of if-then-else rules



THE GINI MATH FOR THE FIRST CUT

Rectangle	Repay	Default	Gini
R1 (credit > 681)	10	3	0.356
R2 (credit <= 681)	2	9	0.298

BEFORE

Gini = 0.500

(max impurity)

AFTER

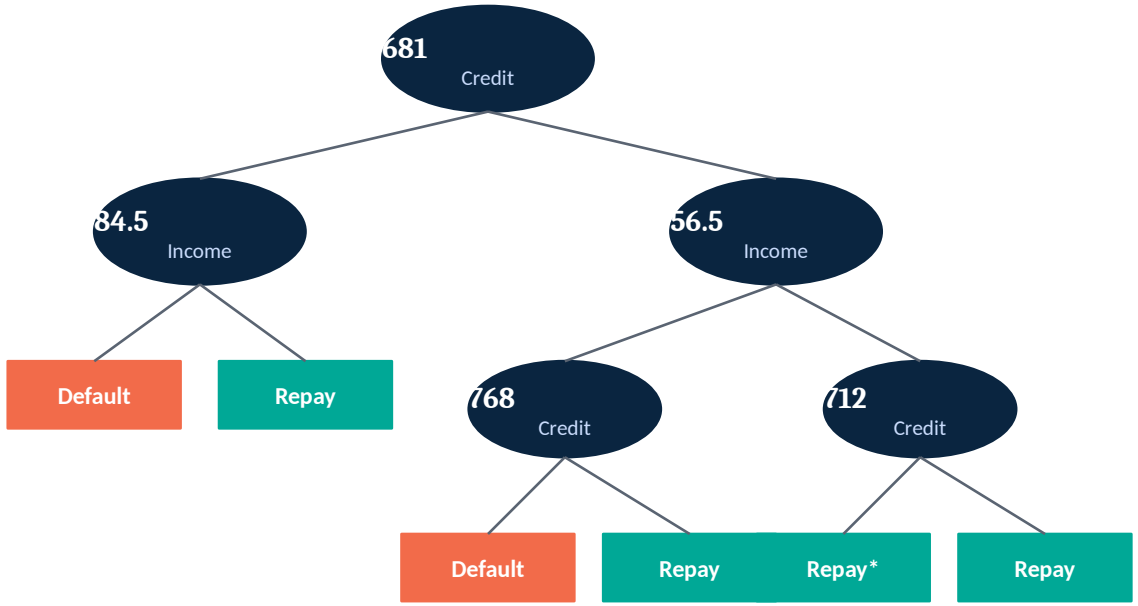
Gini = 0.3286

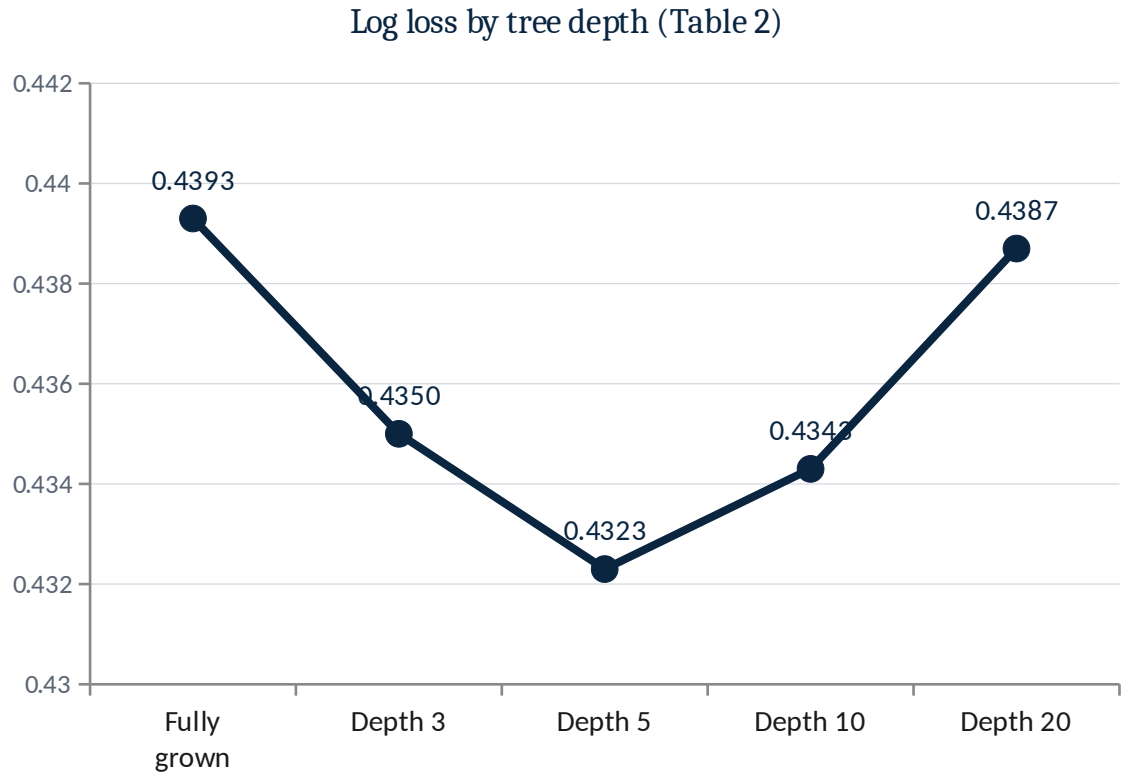
Info gain = 0.1714

Recursive partitioning: at every step, try every feature-and-value pair, pick the one that reduces Gini the most, then repeat on each sub-rectangle. Left alone, the algorithm keeps going until every terminal node holds a single class.

Fully-grown trees overfit. Pruning to depth 5 wins.

THE FULLY-GROWN TREE (FIGURE A-5)





KEY IF-THEN-ELSE RULES

. If 681 < credit < 768 AND income < $56,500 -> **DEFAULT**
. If credit <= 681 AND income < $84,500 -> **DEFAULT**
. Otherwise -> **REPAY**

Winner: Depth 5 . log loss = **0.4323**
Deeper trees overfit training noise; shallower trees underfit real signal.

Random Forest: many weak trees, one strong vote

HOW WE CREATE THE VARIATION

1 Bootstrapping
Each tree sees a different random sample (with replacement) of the 264K observations. Some borrowers appear multiple times, others not at all.

2 Random feature selection
At every node, the algorithm considers only sqrt(p) ~ sqrt(70) ~ 8 randomly-chosen features. No single feature can dominate every tree.

PROVOST AND FAWCETT'S ANALOGY

"Imagine each model as an expert. If every expert had the exact same knowledge, they'd predict the same outcome and additional experts add no value. Because each has a slightly different perspective, collectively they provide more information than any single expert on her own."

HYPERPARAMETER GRID (BY CROSS-VALIDATION)

Number of trees:

150	200	**250**	300

% features per node:

10%	20%	30%	**40%**

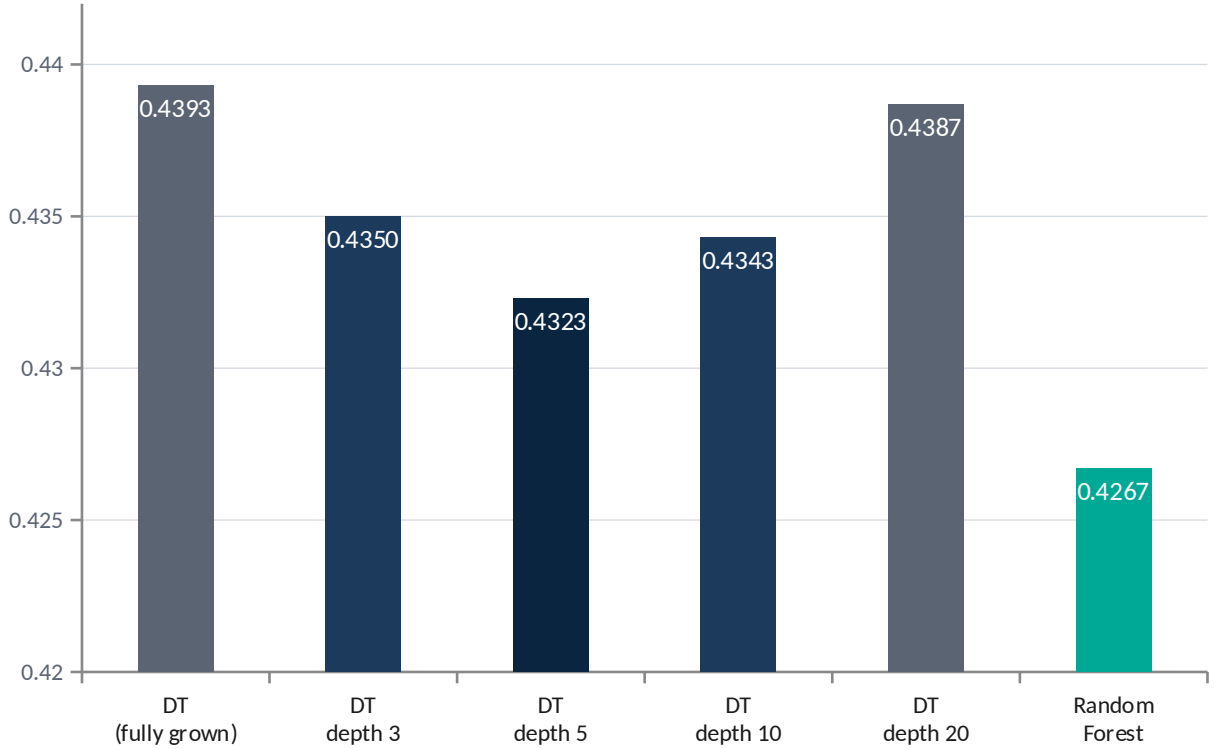
WINNING COMBINATION

250 x 40%

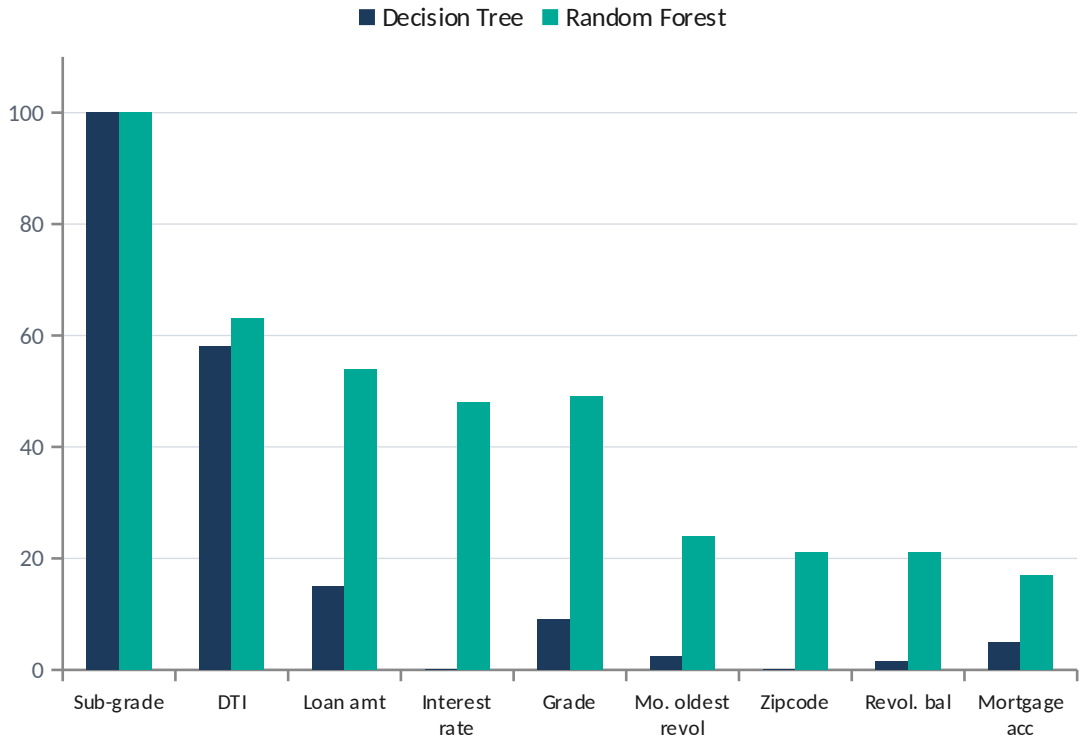
250 bootstrapped trees, each node picks from 40% of the 70 features (~28 per node, well above sqrt(70)).

The showdown: Random Forest wins on log loss

Log loss across all six models



Feature impact (%) . DT vs. RF



Winner: Random Forest . **log loss 0.4267** *(beats depth-5 tree by 0.0056)*

Why RF wins: the tree leans on 4 dominant features; RF recruits 10+ that the tree ignores (interest rate, zipcode, revolving bal) turning weak signals into a stronger vote.

Our recommendation to Figel's wealth-management firm

VERDICT

Selective, capped exposure - yes. *Whole-portfolio bet - no.*

1 Use the RF model as a screen, not the sole signal

Pair the 0.4267-log-loss Random Forest with LC's own grade + a manual credit committee review before funding.

2 Cap allocation at 3-5% of client AUM initially

Consistent with the return profile (~7% net) and the entity + transparency risks the model cannot quantify.

3 Skew to Grade B-C, avoid E and FG

Table 3's net returns are similar (7-8%) but default rates diverge sharply. B/C offer the same reward with less tail risk.

4 Re-underwrite quarterly

Income-verification rates dropped from 49% to 26.8% in 3 years. Any drift in LC's own controls should trigger a stop-loss.

Thank you.

Presented by Group 6

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